

Boundary integral method for Poisson equation with orbital data

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We have developed a boundary integral method to solve the Poisson equation with dielectric interfaces, using orbital data as the source term. In this note, I briefly describe the solver and show some preliminary results.

First, as has been discussed in the previous notes, the equation we solve is

$$\left(\frac{1}{2} \frac{\epsilon_i + \epsilon_j}{\epsilon_i - \epsilon_j} + D^T\right) \sigma(x) = -\partial_{\mathbf{n}} u_{\text{inc}}(x), \quad x \in \partial\Omega_i \cap \partial\Omega_j, \quad (1)$$

where $\sigma(x)$ is the unknown surface charge density, D^T is the transpose of the double layer potential operator, and $u_{\text{inc}}(x)$ is the incident potential generated by the orbital data, given by

$$\partial_{\mathbf{n}} u_{\text{inc}}(x) = \int_{\Omega_o} \frac{\partial G(x, y)}{\partial \mathbf{n}_x} \rho(y) dy, \quad (2)$$

where $G(x, y)$ is the Green's function for the Poisson equation, and $\rho(y)$ is the charge density computed from the orbital data. Equation (1) is a Fredholm integral equation of the second kind¹. Solving it yields the surface charge density $\sigma(x)$, which we then use to compute the scattered potential and field at any point in the domain. Our algorithm consists of the following steps:

1. Load the orbital data and compute the charge density $\rho(y)$ in the orbital region Ω_o
2. Discretize the boundary into a set of panels, and evaluate $\partial_{\mathbf{n}} u_{\text{inc}}(x)$ on each panel
3. Construct the discretized representation of the operator D^T using the discretized boundary
4. Solve the resulting linear system to obtain the surface charge density $\sigma(x)$

We first load the Wannier functions stored in the `.xsf` files. For ease of calculation, we shift each Wannier function to be centered at the atom and truncate it in z , as shown below:

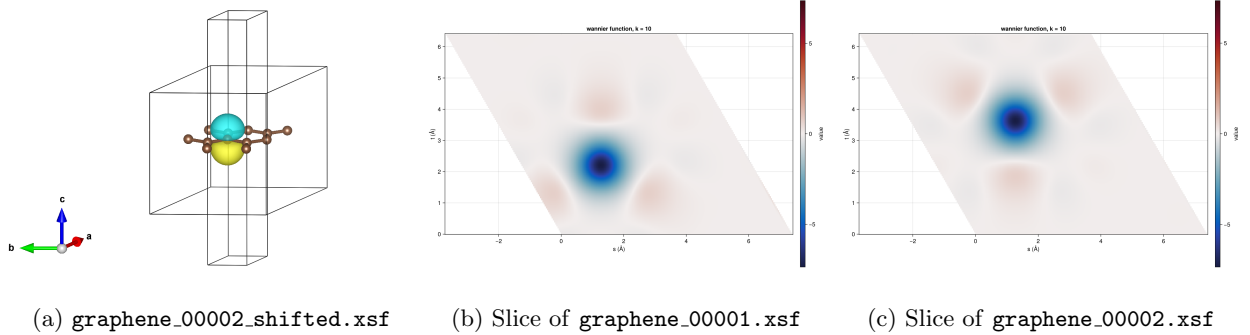


FIG. 1: Wannier functions from the `.xsf` files.

Then, by multiplying the Wannier functions, we obtain the charge density $\rho(y)$. In the following examples, we consider the density generated by the Wannier function in `graphene_00002.xsf` multiplied by itself. We embed this density into a box of size $(90, 90, 9)$ to represent the 2D material, with relative permittivity $\epsilon_r = 6$. Based on the charge density, we discretize the surface into panels, and on each panel we use a $p \times p$ Gauss–Legendre quadrature to sample the right-hand side of Eq. (1). The orbital density and the panels are shown in Fig. 2. Panels near the orbital density and near edges are refined to capture the kernel singularity and the corner singularity, respectively.

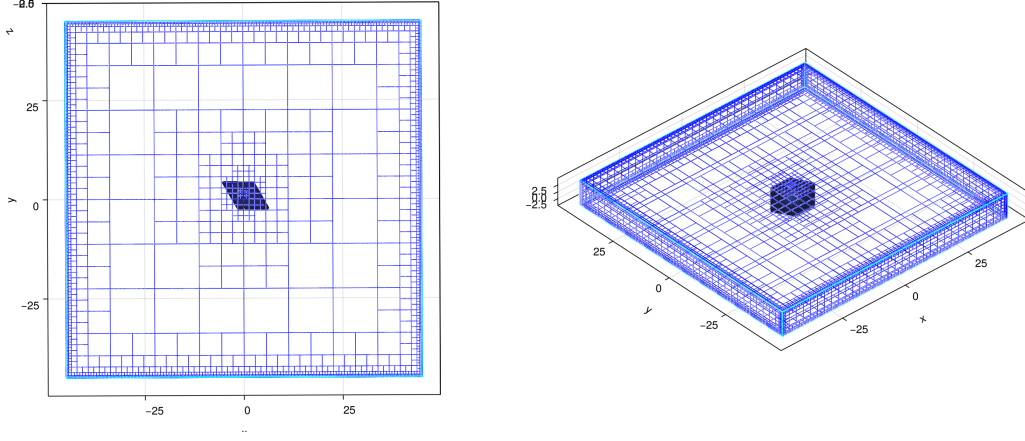


FIG. 2: Orbital density and panels on the boundary of the dielectric interface.

With the discretized surface, we construct a discretized representation of the operator (left-hand side) and the data (right-hand side) in Eq. (1), and solve the resulting linear system to obtain the surface charge density $\sigma(x)$, as shown in Fig. 3.

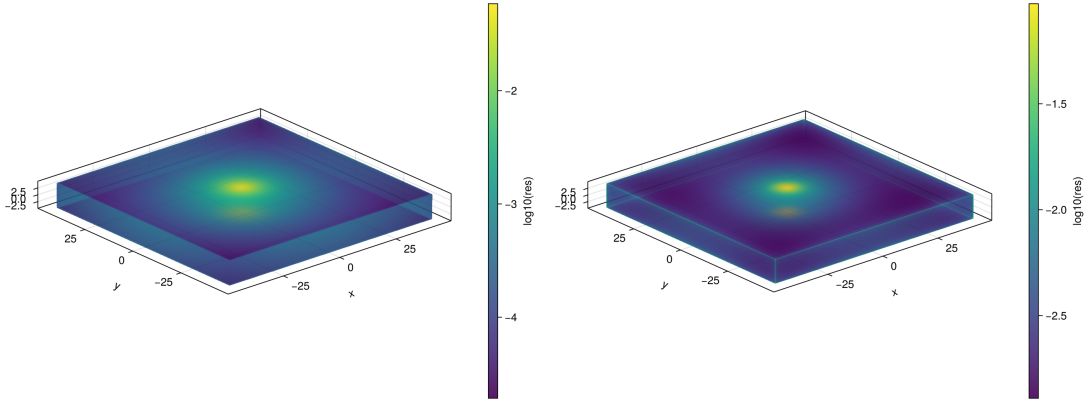


FIG. 3: RHS (left) and solution (right) of Eq. (1).

Finally, with the discretized representation of the surface charge density $\sigma(x)$, we can evaluate the potential at any point in the domain. To validate the solver, we perform a self-convergence test by evaluating the potential at a set of fixed points with different quadrature orders and edge refinements. Figure 4 shows that the potential converges exponentially with the edge-refinement level.

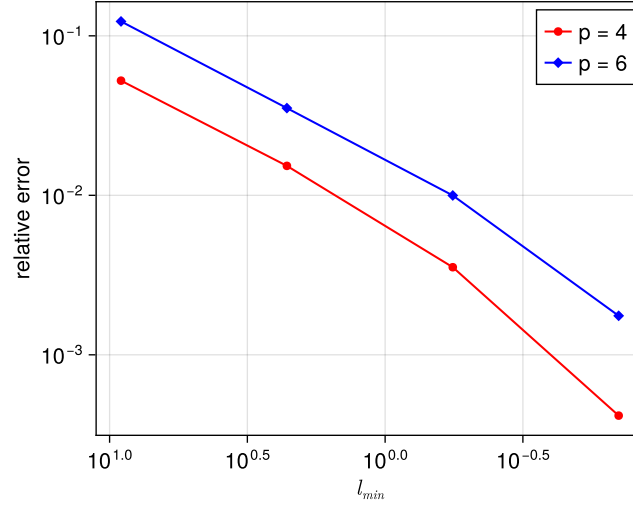


FIG. 4: Convergence of the potential at fixed points with different quadrature orders p and edge refinements l_{min} .

¹ L. Greengard and M. Moura, Acta numerica **3**, 379 (1994).